

# Yanjun Chen 陈彦筠

PhD Candidate · Department of Computing · The Hong Kong Polytechnic University

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## Research Interests

**Reinforcement Learning for Large Language Models and Embodied Agents:** models are trainable; the environments that train them are not. Toward making the environment trainable, the way models are.

## Education

**The Hong Kong Polytechnic University**, Hong Kong Sept 2024 to present

PhD, Department of Computing. Candidature confirmed June 2026.

Advisors: Prof. Wenjie Li (Maggie), Prof. Wei Zhang. Joint doctoral training with the Eastern Institute of Technology (EIT), Ningbo.

**Jinan University**, Guangzhou, China Sept 2020 to Jun 2024

B.Eng. in Cyberspace Security. GPA 88.6 / 100 · Rank 6 of 62 · Outstanding Graduate.

## Publications

### First-Author

- Yanjun Chen**, Yirong Sun, Hanlin Wang, Jinghan Wang, Xinming Zhang, Xiaoyu Shen, Wenjie Li, Wei Zhang. **Exact Is Easier: Credit Assignment for Cooperative LLM Agents**. *arXiv:2603.06859*, 2026 (in submission).
- Yanjun Chen**, Yirong Sun, Xinghao Chen, Jian Wang, Xiaoyu Shen, Wenjie Li, Wei Zhang. **Integrating Chain-of-Thought for Multimodal Alignment: A Study on 3D Vision-Language Learning**. *arXiv:2503.06232*, 2025. [dataset]
- Yanjun Chen**, Xinming Zhang, Xianghui Wang, Zhiqiang Xu, Xiaoyu Shen, Wei Zhang. **Rethinking Soft Actor-Critic in High-Dimensional Action Spaces: The Cost of Ignoring Distribution Shift**. *arXiv:2410.16739*, 2024.
- Yanjun Chen**, Dawei Zhu, Yirong Sun, Xinghao Chen, Wei Zhang, Xiaoyu Shen. **The Accuracy Paradox in RLHF: When Better Reward Models Don't Yield Better Language Models**. *EMNLP 2024*.  
*One further first-author manuscript under review, 2026.*

### Co-Authored

Nine co-authored papers, including *EMNLP 2025*, *Findings of ACL 2025*, and *NAACL 2025*; full list on [Google Scholar](#).

## Open Source

**C3** [github.com/EIT-EAST-Lab/C3](https://github.com/EIT-EAST-Lab/C3)

Official implementation of *Exact Is Easier* (arXiv:2603.06859): exact per-decision credit via transcript replay, with a method-agnostic audit of credit quality.

**AccuracyParadox-RLHF** [github.com/EIT-NLP/AccuracyParadox-RLHF](https://github.com/EIT-NLP/AccuracyParadox-RLHF)

Official implementation of *The Accuracy Paradox in RLHF* (EMNLP 2024): reproducible training pipelines, evaluation harnesses, and reference reward models.

**3D-CoT dataset** [huggingface.co/datasets/Battam/3D-CoT](https://huggingface.co/datasets/Battam/3D-CoT)

Chain-of-thought annotations for 3D vision-language alignment (arXiv:2503.06232).

**Myco**, independent project [github.com/Battam1111/Myco](https://github.com/Battam1111/Myco)

Agent infrastructure in Rust: persistent memory and knowledge management for AI agents in the Model Context Protocol (MCP) ecosystem.

## Technical Skills

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|----------------|------------------------------------------------------------------------------------------------------------------------------|
| Post-Training  | SFT and RLHF; PPO, DPO, GRPO; reward modeling and evaluation; RLVR; SAC and off-policy RL for continuous control.            |
| Training Infra | DeepSpeed (ZeRO), Megatron-LM; verl, OpenRLHF, Hugging Face TRL; LoRA / PEFT; multi-node multi-GPU training; vLLM inference. |
| Frameworks     | PyTorch, Hugging Face Transformers / Datasets / Accelerate; Weights & Biases.                                                |
| Programming    | Python, C/C++, Bash; Git, Linux.                                                                                             |
| Languages      | Mandarin (native), English (professional), Japanese (elementary).                                                            |

## Teaching & Service

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|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Teaching (TA) | COMP5221 Software Project Management · COMP1002 Computational Thinking and Problem Solving · COMP1411 Introduction to Computer Systems · COMP5567 Distributed Algorithms and Protocols for Blockchains (2024 to 2026). |
| Reviewing     | ACL Rolling Review (ARR) 2026 · IEEE Transactions on Neural Networks and Learning Systems (TNNLS).                                                                                                                     |